

Lecture 11: Early CAPM Tests and First Cracks

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Intro

- We now start the factor-model module in empirical asset pricing
- Today is about the first CAPM tests and the first cracks in the model
- Logic: CAPM benchmark first, data second, anomalies third, joint tests last

- Part I: why CAPM looked compelling in theory ([Sharpe 1964](#); [Lintner 1965](#); [Mossin 1966](#))
- Part II: early empirical protocols from Black, Jensen, and Scholes ([1972](#)) and Fama and MacBeth ([1973](#))
- Part III: the first anomaly evidence ([Basu 1977](#); [Banz 1981](#); [De Bondt and Thaler 1985](#))
- Part IV: joint testing logic, from Roll ([1977](#)) to Gibbons, Ross, and Shanken ([1989](#))

Part I: CAPM Recall

- In 1952: Markowitz brings us to light with “Modern Portfolio Theory”
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In the 1960s:

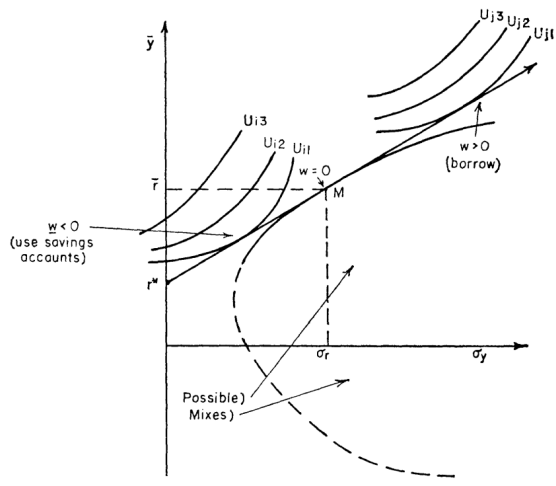
- Sharpe (1964), Lintner (1965), and Mossin (1966) build on Markowitz to get the CAPM
- Notice this way before the Rational Expectation revolution and the SDF framework
- These guys were The Beatles of Finance!

What CAPM Delivered

- It linked portfolio choice to equilibrium expected returns
- It delivered one sufficient statistic: market beta
- Investors are mean-variance optimizers over one period: you like returns, dislike variance
- Homogeneous beliefs and frictionless trading are assumed
- Everyone can borrow/lend at one risk-free rate and hold the same assets
- Important: different investors will have different preferences!

Mean-Variance Frontier and Tangency Portfolio

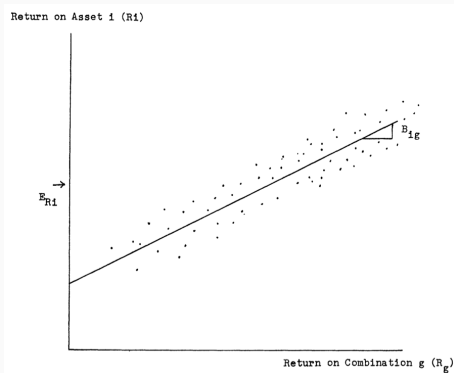
- Efficient portfolios are combinations of the risk-free asset and *one* tangency portfolio
- Tangency portfolio: highest Sharpe-ratio risky portfolio
- Investors differ only in how much risk they scale, not in risky composition
- They will *not* have the same overall portfolio!



Security Market Line

- Expected excess returns are linear in market beta
- Only systematic market covariance is priced
- Idiosyncratic risk earns no premium

$$\mathbb{E}[R_i - R_f] = \beta_i \mathbb{E}[R_m - R_f], \quad \beta_i \equiv \frac{\text{cov}(R_i, R_m)}{\text{var}(R_m)}$$



CAPM as SDF/Beta Representation (Bridge to Lecture 4)

$$m_{t+1} = a - bR_{m,t+1}$$

- CAPM is one special case of the general SDF framework from Lecture 4
- If the SDF is affine in market returns, beta pricing follows mechanically
- This gives a clean bridge between no-arbitrage language and equilibrium CAPM
- The idea of Bansal and Yaron ([2004](#)) to think about a “wealth portfolio” is much older than you think!

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_i(R_{m,t} - R_{f,t}) + \varepsilon_{i,t}$$

- Cross-section: higher beta assets should have higher average returns
- The relation should be approximately linear
- Exact CAPM implies $\alpha_i = 0$ for all test assets
- Nonzero alpha patterns indicate systematic model failure
- Important: nothing else should be added to the right-hand side!

Questions?

Black (1972): Zero-Beta CAPM

$$\mathbb{E}[R_i] = \mathbb{E}[R_z] + \beta_i(\mathbb{E}[R_m] - \mathbb{E}[R_z])$$

- Black (1972) relaxes unrestricted risk-free borrowing and lending
- A zero-beta portfolio replaces R_f as the intercept object
- The linear beta-pricing equation survives with a different baseline return

Zero-Beta CAPM: Derivation Intuition

$$\text{cov}(R_z, R_m) = 0, \quad \beta_i \equiv \frac{\text{cov}(R_i, R_m)}{\text{var}(R_m)}$$

- Without risk-free borrowing/lending, the efficient set is generated by risky portfolios only
- The relevant baseline return is the expected return on the portfolio orthogonal to R_m
- Orthogonality gives the same slope object (β_i), but a different intercept
- Intercept shifts from R_f to $\mathbb{E}[R_z]$

Zero-Beta Assets Are Cool Again

- The interest rates we observe on the data are a mix of many things
- Policy decisions + expectations about the future + risk premia + market frictions
- In the CAPM world, all we need is a zero-beta asset to get the main implications...

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Even more recently:

- Kocherlakota ([2025](#)) analyzes the theoretical price of sovereign debt discount by the zero-beta rate

Questions?

Part II: Early CAPM Tests

From Theory to Data: What Early Tests Actually Did

- Black, Jensen, and Scholes ([1972](#)) decided to “draw” the Security Market Line
- Do high-beta stocks earn higher average returns?
- Is this increase proportional to the market risk premium?
- By the way, “return on wealth” became “broad equity index return” really fast (see Roll ([1977](#)))

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- First empirical issue: how to reliably measure β_i ? Returns are noisy!
- Solution: group stocks into portfolios sorted on *past betas* to reduce noise

Black-Jensen-Scholes (1972): Design + Regression Setup

- In a given time t , for N stocks, estimate $[\beta_1, \dots, \beta_N]$:

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_i(R_{m,t} - R_{f,t}) + \varepsilon_{i,t}$$

- Create 10 portfolios sorted on individual past betas
- Hold each portfolio for one year and redo sorting process every year
- At the end, you have 10 time series of portfolio returns

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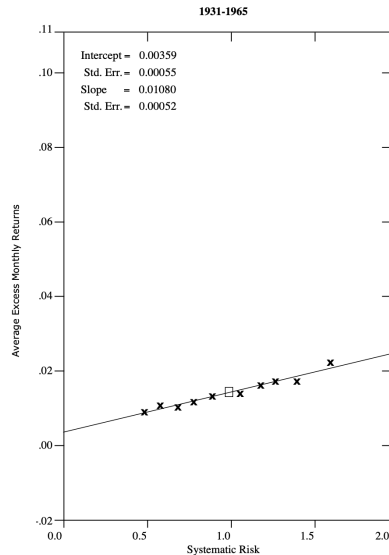
- Create 10 portfolios sorted on individual past betas
- Hold each portfolio for one year and redo sorting process every year
- At the end, you have 10 time series of portfolio returns
- Redo the same regression for each of the 10 portfolios to get 10 alphas and 10 betas
- Finally: estimate γ_0 and γ_1 from:

$$\mathbb{E}[R_{p,t} - R_{f,t}] = \gamma_0 + \gamma_1\beta_p + u_{p,t}$$

- Important: this is a **cross-sectional** regression!

BJS Main Findings

- Beta and average returns are positively related
- Low-beta portfolios earn “too much”
- High-beta portfolios earn “too little”
- $\hat{\gamma}_0 > 0$, $\hat{\gamma}_1$ is small



BJS Regression Evidence

	<i>Time Period</i>				
	<i>Total Period</i>	<i>Subperiods</i>			
	1/31–12/65	1/31–9/39	10/39–6/48	7/48–3/57	4/57–12/65
$\hat{\gamma}_0$	0.00359	-0.00801	0.00439	0.00777	0.01020
$\hat{\gamma}_1$	0.0108	0.0304	0.0107	0.0033	-0.0012
$\gamma_1 = \bar{R}_M$	0.0142	0.0220	0.0149	0.0112	0.0088
$t(\hat{\gamma}_0)$	6.52	-4.45	3.20	7.40	18.89
$t(\gamma_1 - \hat{\gamma}_1)$	6.53	-4.91	3.23	7.98	19.61

- This was taken as a partial victory: higher beta, higher returns
- The relationship is flatter than expected...
- This is in fact support for the zero-beta CAPM!

- Frazzini and Pedersen (2014) showed this “anomaly” is still alive, and true across international markets
- Also true in bond markets and commodities!
- They provide a model and a theory to explain it: embedded leverage

- Frazzini and Pedersen (2014) showed this “anomaly” is still alive, and true across international markets
- Also true in bond markets and commodities!
- They provide a model and a theory to explain it: embedded leverage
- Some investors cannot lever up... high-beta stocks provide leverage
- They bid up the prices, so returns tend to be lower than CAPM would predict
- Fun fact: they show how Warren Buffet has been buying low-beta stocks for decades!

Questions?

Fama-MacBeth (1973): Why Two Passes?

- Exposure estimation and risk-price estimation are different statistical problems
- Two-pass procedure separates those two steps cleanly ([Fama and MacBeth 1973](#))
- It produces a time series of risk-price estimates for inference
- This became the default empirical workflow in cross-sectional asset pricing

Fama-MacBeth Pass 1 (Estimate Betas)

$$R_{i,t} - R_{f,t} = a_i + b_i(R_{m,t} - R_{f,t}) + e_{i,t}$$

- First-pass time-series regressions estimate each asset's market beta
- Betas are treated as characteristics for the second-pass cross section
- Portfolio grouping can be used too (in fact used in Fama and MacBeth (1973))

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Pass 2:

$$R_{i,t} = \gamma_{0,t} + \gamma_{1,t}\hat{\beta}_i + u_{i,t}$$

- At each date, run a cross-sectional regression of returns on estimated betas
- This will deliver two time series of estimates: $\{\hat{\gamma}_{0,t}\}$ and $\{\hat{\gamma}_{1,t}\}$
- Average $\gamma_{1,t}$ over time to estimate the market price of beta risk
- Average $\gamma_{0,t}$ to assess the intercept restriction

- A positive average beta-return relation is present
- Intercept evidence is not cleanly aligned with strict Sharpe-Lintner CAPM
- Still the backbone workflow, but with generated-regressor and beta-estimation caveats
- Conclusions depend on test assets, sample period, and market proxy quality
- [Click here for huge tables](#)
- The Roll (1977) critique is always hunting us, by the way ...

If beta is enough, why did anomalies show up so quickly?

Part III: The First Cracks

Basu (1977): Characteristics vs Beta

- Basu (1977) sorts stocks by earnings-to-price ratios
- High E/P portfolios earn higher average returns than low E/P portfolios
- This spread remains difficult to map into a pure one-beta CAPM story
- Early signal that characteristics may contain pricing information
- CAPM says beta should span expected-return differences; Basu says E/P also forecasts returns

Banz (1981): Size Effect

- Banz (1981) documents higher average returns for small-cap stocks
- The size-return spread is hard to eliminate with market beta alone
- Small-minus-big logic later becomes a canonical factor construction
- Another direct empirical crack in single-factor CAPM

DeBondt-Thaler (1985): Long-Run Reversal

- De Bondt and Thaler (1985) show that long-horizon losers tend to outperform winners
- Multi-year return reversals are difficult for static CAPM to rationalize
- Short takeaway for this lecture: return dynamics matter for pricing tests
- Behavioral interpretations enter the CAPM debate more directly

What These Cracks Have in Common

- They generate systematic nonzero pricing errors relative to CAPM
- Firm characteristics and past returns carry incremental information
- One-factor beta seems too narrow for the observed cross section
- The field now needs stronger model-comparison tools

Early Anomalies (Figure/Table Placeholder)

[Insert anomaly spreads figure/table here]

Part IV: Joint Testing and Synthesis

Why We Need Joint Tests (Roll-to-GRS Motivation)

- Testing each asset separately creates a multiple-testing problem
- A valid model should price a set of assets jointly, not one by one
- Roll (1977): every CAPM test is joint with the chosen market proxy
- The natural next step is a formal joint test on the alpha vector

$$R_t^e = \alpha + Bf_t + \varepsilon_t, \quad t = 1, \dots, T$$

$$H_0 : \alpha = 0_N$$

- Gibbons, Ross, and Shanken ([1989](#)) propose an exact finite-sample test of this joint null
- Rejection means the tested factor portfolio is not mean-variance efficient for that asset set
- Notation: N test assets, K factors, T time observations

$$GRS = \frac{T - N - K}{N} \left(1 + \hat{\mu}_f' \hat{\Sigma}_f^{-1} \hat{\mu}_f \right)^{-1} \hat{\alpha}' \hat{\Sigma}_\epsilon^{-1} \hat{\alpha} \sim F_{N, T-N-K} \text{ under } H_0$$

- $\hat{\alpha}$: vector of estimated pricing errors from the time-series regressions
- $\hat{\Sigma}_\epsilon$: residual covariance matrix across test assets
- $\hat{\mu}_f, \hat{\Sigma}_f$: sample mean and covariance of factor returns

GRS Practical Insight + Figure Space

- Good models need jointly small alphas, not only scattered insignificant t-stats
- Statistical rejection and economic materiality are different objects
- Conclusions depend on which assets and factors are included in the test
- GRS remains a core diagnostic for factor-model performance

[Insert GRS p-value / alpha heatmap figure here]

- CAPM gave a disciplined starting benchmark
- BJS and Fama-MacBeth established the empirical playbook
- Basu, Banz, and DeBondt-Thaler revealed systematic cracks
- GRS formalized joint model evaluation in finite samples
- Next step: the Fama-French revolution and the multi-factor era

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